

Identifying Underserved Areas for Park Development in Hamilton City Using GIS

Research Question & Importance

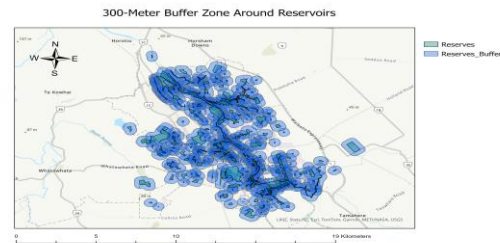
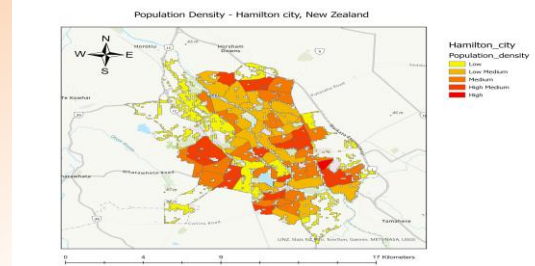
How can GIS be used to identify underserved residential areas in Hamilton City that require new parks?

Urban green spaces are essential for public health, social well-being, and urban resilience. Ensuring all residents have easy access to parks supports equity and sustainability.

Literature Review

Urban green spaces play a crucial role in enhancing city life by offering recreational areas, improving mental health, reducing urban heat, supporting biodiversity, and strengthening communities. As Hamilton grows, equitable access to these spaces is essential for sustainable development. Research by Wolch et al. (2014) and Kabisch and Haase (2014) highlights green space inequity, where wealthier areas have better access than underserved communities. They advocate using GIS tools to identify and address these gaps. In New Zealand, policies like the Ministry for the Environment's "Our Land 2021" report stress the importance of integrating green infrastructure, especially in high-density areas.

GIS enables planners to analyze spatial patterns and identify where green spaces are most needed. Techniques such as buffer and network analysis help detect underserved areas and guide the placement of new parks. Zhang et al. (2011) show how GIS reveals disparities not easily seen through traditional methods. In Hamilton, where urban intensification may reduce open space availability, GIS supports strategic planning by considering population density, land use, and existing infrastructure. This project builds on these insights to develop a GIS-based approach for locating new parks in underserved, high-density parts of the city.

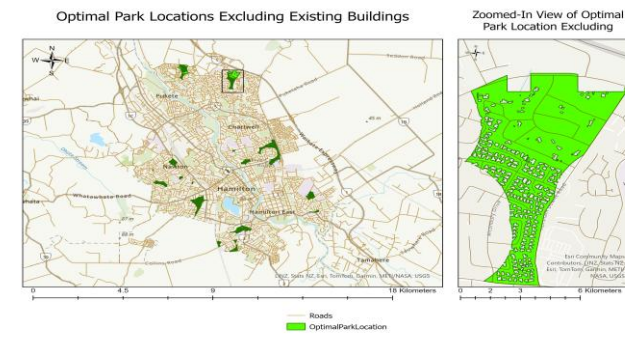


Data and Analysis

- Population, building, road, and park layers were imported.
- Population density was calculated and symbolized.
- Parks were buffered by 300m.
- Areas not covered were extracted using Erase tool.
- High-density zones were selected and intersected with underserved areas.
- Building footprints were erased from the output.
- Result shows proposed park zones in needful areas.

Results and Interpretation

The final map shows 17 suburbs as underserved. These areas have high population densities and lack parks within 300m.



Optimal Park Locations

OBJECTID	SA2(Suburb)	Count
1	Crawshaw	1
2	Deanwell	1
3	Enderley North	1
4	Enderley South	1
5	Fairfield North (Hamilton City)	1
6	Fairfield South (Hamilton City)	1
7	Flagstaff North East	1
8	Hillcrest East (Hamilton City)	1
9	Hillcrest West (Hamilton City)	1
10	Kahikatea	1
11	Melville North	1
12	Nawton East	1
13	Nawton West	1
14	Rototuna East	1
15	Rototuna West	1
16	Swarbrick	1
17	Whitiora	1

Limitations & Improvements

The GIS approach effectively identified gaps in park coverage but had limitations. Euclidean buffers may overestimate access by ignoring street networks, and population data might be outdated. Land ownership, ecological factors, and social or economic criteria were not included. Still, the analysis offers a solid starting point for more detailed planning.

Conclusion

GIS effectively identified underserved, high-density areas suitable for new parks. This approach supports equitable urban planning.

References

- Kabisch, N., & Haase, D. (2014). Green justice or just green? *Landscape and Urban Planning*, 122, 129–139.
- Maas, J., Verheij, R. A., Groenewegen, P. P., de Vries, S., & Spreeuwenberg, P. (2006). Green space, urbanity, and health: How strong is the relation? *Journal of Epidemiology & Community Health*, 60(7), 587–592
- Ministry for the Environment. (2021). *Our land 2021*. New Zealand Government.
- Wolch, J. R., Byrne, J., & Newell, J. P. (2014). Urban green space, public health, and environmental justice: The challenge of making cities "just green enough." *Landscape and Urban Planning*, 125, 234–244.
- Zhang, X., Lu, H., & Holt, J. B. (2011). Modeling spatial accessibility to parks: A national study. *International Journal of Health Geographics*, 10, 31.